

Topology-Enhanced Multi-Modal Residual Graph Convolutional Network for Autism Spectrum Disorder Identification

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Abstract—Functional Brain Networks (FBNs) have shown promise in diagnosing neurological disorders, particularly Autism Spectrum Disorder (ASD). Currently, Graph Convolutional Networks (GCNs) excel in processing FBN data, making them instrumental in learning from functional Magnetic Resonance Imaging (fMRI) and analyzing ASD. Despite their advantages, existing methods primarily concentrate on the structural properties of brain networks, often overlooking the essential topological information. To bridge this gap, we introduce a Topologically Enhanced Multi-Pattern Residual Graph Learning (TEMPRGL) framework. Our approach begins by constructing diverse brain networks from fMRI data, capturing a range of connectivity patterns. We then employ a dual-feature extraction strategy on FBNS: one that utilizes multiple Residual GCN modules to generate multi-modal graph representations, which are subsequently integrated; and another focuses on combining topology information from FBN. To validate our TEMPRGL, we experiment on the Autism Brain Imaging Data Exchange (ABIDE) dataset. It attained an impressively high accuracy rate of 92.5%, underscoring its effectiveness in neurological disorder diagnosis.

Index terms—Autism Spectrum Disorder, Graph Convolutional Neural Networks, Functional Connectivity Networks, Topology

I. INTRODUCTION

Autism Spectrum Disorder (ASD) is a condition related to the development of the nervous system, marked by repetitive or stereotypical behaviors, social cognition impairments, and communication impairments [1, 2]. Despite the unknown cause, early and appropriate interventions show promise in improving social cognition and communication impairments

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[3]. Nevertheless, the absence of biomarkers makes early ASD diagnosis challenging, relying on symptom-based identification [4]. Hence, urgently needed is a prompt and accurate diagnostic system for ASD. Advances in MRI technology have enabled the creation of Functional Brain Networks (FBNs) through resting state functional Magnetic Resonance Imaging (fMRI), providing a valuable approach to detect the nuanced brain changes characteristic of ASD[5, 6]. FBN, unveiled through correlations in nervous system activities, are acknowledged as fundamental organizational elements in the human brain. The construction of FBN entails the creation of a network by modeling the patterns of functional associations among brain regions. In this network, nodes are assigned to Regions Of Interest (ROIs) within the brain, while edges depict the functional connections linking two ROIs [7, 8]. Popular methods for constructing FBN include correlation-based approaches [9], Granger causality analysis [10], and others.

Currently, there has been a notable surge in the adoption of Graph Neural Networks (GNNs) for holistic graph-based learning endeavors. This trend is particularly evident in the realm of advanced data analysis, where GNNs have been instrumental in achieving state-of-the-art outcomes in the examination of graph-structured data [11, 12]. This trend has extended to the field of fMRI analysis, where some researchers have successfully applied GNN [13, 14]. While many have primarily employed single connectivity patterns for analysis, recognizing the brain's complexity, some researchers are exploring multiple connectivity patterns to harness potential complementary information and address the limitations of singular approaches [15-17]. Nevertheless, these approaches tend to concentrate on the structural aspects of brain networks, often overlooking the crucial topological details embedded within [18].

To tackle this concern, we propose the Topologically Enhanced Multi-Pattern Residual Graph Learning (TEMPRGL) framework. As depicted in Fig. 1, we first obtain the FBN from fMRI data by using Pearson correlation [19, 20], partial sparse representation correlation [21-23] and

Granger causality [24]. Then, drawing inspiration from specific FBNs, we perform two distinct types of feature extraction on the FBN [18]. One involves employing multiple Residual Graph Convolutional Network (GCN) modules to obtain multi-modal graph representations for each theme, followed by their fusion [25, 26]. The other focuses on extracting a vectorized representation of the FBNs information. Subsequently, these features are merged, and the fused characteristics are transmitted via fully connected layers. The final output of the fully connected layer is then used for disease classification, enabling effective diagnostic outcomes.

In this study, we primarily achieved the following objectives:

1. Based on the information available to us, the introduction of specific FBNs topological information within various connectivity patterns is a novel approach. Furthermore, experimental results unequivocally demonstrate that the integration of this specific FBN information effectively compensates for the information gaps resulting from the modeling of brain networks.
2. The utilization of a two-layer GCN with residual connections allows our graph representation learning to balance the capability of each node in aggregating neighbor information. Experimental results indicate that the addition of residual connections enhances the accuracy of ASD classification.

II. RELATED WORKS

A. ASD Identification

Recent studies on ASD identification are converging into two primary methodologies: (1) conventional machine learning techniques and (2) deep learning innovations. Conventional strategies often rely on the integration of handcrafted rs-fMRI features, such as nodal degree and clustering coefficients, into established classification frameworks, with models such as Support Vector Machines. The emergence of deep learning has ushered in a wave of novel diagnostic approaches leveraging rs-fMRI data. Park and colleagues presented a deep learning framework that employs residual networks and attention mechanisms to extract spatial and temporal characteristics from 4D fMRI images of individuals diagnosed with ASD [14]. Fang et al. have developed UFA-Net, an unsupervised cross-domain fMRI adaptation framework, marking the first instance of its kind aimed at the automatic identification of depression through cross-site fMRI data [27]. However, they only considered one FBNs construction method, potentially ignoring the brain's underlying deep information.

In this paper, we focus on leveraging both node and graph information for ASD identification.

III. DATA PROCESSING AND METHODOLOGY

A. Data Gathering

In our research, we utilized the identical dataset referenced in Study [28]. This dataset contains resting-state functional magnetic resonance imaging (rs-fMRI) data from 92 participants, including 47 normal developing individuals aged 7 to 15 years as the normal control (NC) group, and 45 individuals with autism spectrum disorder (ASD). The Pre-treatment of the data was carried out using the Statistical Parametric Mapping software (SPM8). During this process, the first 10 volumes from each participant are discarded to reduce signal artifacts. The subsequent images underwent several preprocessing steps: (1) normalization to Montreal Neurological Institute (MNI) space at a resolution of $2 \text{ mm} \times 2 \text{ mm} \times 2 \text{ mm}$; (2) regression of nuisance signals, including those from ventricles, white matter, global signals, and head motion, employing Friston's 24-parameter model [29]; (3) implementation of band-pass filtering in a frequency band of 0.01-0.08 Hz; and (4) removal of trends from the signal. Finally, we extracted the preprocessed ROI data based on the Anatomical Automatic Labeling (AAL) atlas $x \in R^{90 \times 180}$.

B. Brain Network Construction

Subsequently, we use the Pearson correlation to build functional connectivity networks, partial correlation (sparse representation with sparse constraints), and Granger causality as follows:

(1) Pearson's Correlation

The Pearson Correlation (PC) is commonly employed in constructing FBN due to its efficacy in capturing comprehensive correlations between ROIs. Its mathematical representation can be expressed as follows:

$$W_{ij} = \frac{(x_i - \bar{x}_i)^T (x_j - \bar{x}_j)}{\sqrt{(x_i - \bar{x}_i)^T (x_i - \bar{x}_i)} \sqrt{(x_j - \bar{x}_j)^T (x_j - \bar{x}_j)}} \quad (1)$$

(2) Sparse Representation

Pearson's correlation coefficient is commonly used to assess the overall correlation among ROIs [15, 17]. However, its accuracy can be compromised by confounding effects. In contrast, estimation of FBNs based on partial correlations eliminates the influence of confounding variables, addressing this concern. However, solvers for partial correlations frequently face ill-posed issues. As a result, we add a sparse regularization (SR) term as a constraint. When estimating functional brain networks (FBNs) through partial correlations:

$$W = \arg \min_W \sum_{i=1}^n \left\| x_i - \sum_{j \neq i} W_{ij} x_j \right\|^2 + \sum_{j \neq i} |W_{ij}|, \quad (2)$$

the hyperparameter λ regulates the trade-off between the two components.

(3) Granger Causality

While PC and SR assess an undirected network, Granger causality delves deeper to scrutinize the causal dynamics specifically between these ROIs. Specifically, Granger causality assesses the capability of a preceding time series in one ROI to predict future values of a delayed time series in another ROI. Consequently, it constructs an asymmetrical

brain network graph. To elaborate, given two time series $x_i[n]$ and $x_j[n]$, the Granger causality-based functional brain network is defined as follows:

$$W_{ij} = G_{x_i \rightarrow x_j} = \ln \frac{|\Sigma(\zeta_i)|}{|\Sigma(\zeta_j)|}, \quad (3)$$

where ζ_i and ζ_j represent the differences between the restricted and unrestricted regression models(w/o $x_j[n]$), respectively, and $\Sigma(\cdot)$ indicates the variance.

C. Graph Convolutional Neural Network Construction

We will conduct a dual-perspective feature extraction from the three types of FBN identified. The first approach concentrates on the structural information of the networks, capturing the inherent layout and connectivity patterns. The second approach delves into the topological information, exploring the network's global properties and the relationships between its components. Upon acquiring these distinct sets of data, we will synthesize them to form a comprehensive feature set. This integrated feature set will then be subjected to a classification process using fully connected layers, ensuring a robust and multifaceted analysis as shown in Fig. 1.

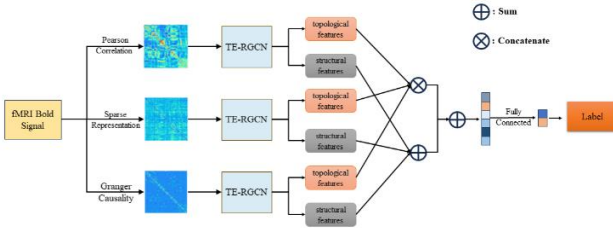


Fig 1. The entire framework of our proposed TEMPRGL

(1) Structural information branch

The brain's FBN can be construed as an irregular graph architecture. Therefore, we can utilize Graph Convolutional Networks (GCNs) proposed by Kipf et al. to extract features. The node update formula is as follows:

$$H^{l+1} = \sigma \left(\widehat{D}^{-\frac{1}{2}} \widehat{A} \widehat{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right), \quad (4)$$

where H denotes the characteristics of nodes in the graph, while W represents the network parameters to be acquired. Additionally, $\widehat{A} = A + I_n$, $\widehat{D}_{ii} = \sum_j \widehat{A}_{ij}$, and $\sigma(\cdot)$ signifies a non-linear activation function.

Our proposed Graph Convolutional Network (GCN) model employs dual-layer convolutions for every configuration graph, utilizing the ReLU activation function in each convolutional layer. To mitigate excessive smoothing of node features, we apply a residual structure. The formula governing the overall forward propagation is delineated as follows:

$$f(X, A) = \text{ReLU}[\widehat{A} \text{ReLU}(\widehat{A} X W^{(0)}) W^{(1)}] + \text{ReLU}(\widehat{A} X W^{(0)}), \quad (5)$$

where $f(X, A) \in R^{n \times d}$, X represents the features of each node (ROI), while A represents the adjacency matrix, which

establishes the relationships among nodes/ ROIs. The weight parameter matrix $W^{(0)}$ denotes the trainable parameters of the initial convolutional layer, whereas $W^{(1)}$ denotes the matrix of weight parameters of the second convolution layer.

We utilize graph structures and content information of the node from diverse modalities of the identical subjects as inputs. We separately train GCN models for diverse modalities to predict the overall graph's category labels. To obtain feature representations for the entire graph, we employ a readout layer to extract the graph-level representation. The readout layer we employ is defined as follows:

$$F = \frac{1}{n} \sum_{i=1}^N f_i(X, A) \parallel \max_{i=1}^N f_i(X, A). \quad (6)$$

As depicted in Fig. 1, we can finally extract vectors representing three different modes: S1, S2, and S3. Subsequently, we concatenate them together, defining it as the structural vector S .

(2) residual connection

In the structural information feature extraction of FBN, multiple GCN modules are used to obtain the multimodal graph representation of each topic. The application of residual network effectively avoids the problems of gradient disappearance and gradient explosion, and provides better classification performance for the model. Although this will increase the cost of computational performance to a certain extent, the experimental results show that the benefits brought by the application of residual structure are considerable

(3) Topology information branch

Drawing inspiration from the method proposed by Zhang et al., we implement their approach to extract topological information through specific FBN extraction techniques [18]. Initially, we flatten the upper triangular part of the FBN into a vector, and then encode this vector using an Multilayer Perceptron(MLP). Its mathematical representation is as follows:

$$F = \text{MLP} \left(\text{Flatten}(\text{Triu}(A)) \right). \quad (7)$$

we ultimately obtain topological representation vectors for three different modes: T1, T2, and T3. They share the same dimensions as S . Subsequently, these three vectors are summed together to form a composite vector defined as the topological vector T .

(4) Fusion and classification

In this section, we integrate two types of information for classification. Firstly, we combine the structural and topological vectors to obtain the final features. Subsequently,

TABLE I. The experimental results in different methods.

Methods	Acc(%)	Precision(%)	Recall(%)	F1score(%)	AUC(%)
PC	83.2±18.7	82.6±22.2	90.3±23.8	84.9±22.4	87.6±2.6
SR	76.9±23.0	77.8±27.9	77.7±31.1	76.4±28.6	83.8±2.5
GCM	74.4±23.7	80.3±25.0	72.9±32.4	73.8±28.2	80.3±2.7
PC_SR	90.4±17.4	83.7±32.7	81.5±34.3	81.7±33.6	92.8±2.7
PC_GCM	87.2±20.2	89.8±23.1	85.4±28.2	86.0±25.9	94.1±2.8
SR_GCM	81.9±22.7	86.1±22.6	81.4±30.1	81.3±27.2	90.7±2.1
PC_SR_GCM	92.5±14.8	94.1±11.8	93.8±15.9	93.4±13.7	96.2±2.3

Notes: The boldface denotes the best results.

TABLE II. Ablation experimental results under different module combinations.

Methods	Acc(%)	Precision(%)	Recall(%)	F1score(%)	AUC(%)
Model 1	90.4±10.4	98.2±5.9	84.7±17.0	89.8±12.3	97.3±1.1
Model 2	91.6±8.8	98.4±5.6	86.4±14.0	91.3±9.8	97.5±1.1
Model 3	91.2±18.3	90.7±23.1	91.8±24.4	90.2±23.8	97.7±1.5
Model 4	92.5±14.8	94.1±11.8	93.8±15.9	93.4±13.7	96.2±2.3

these features are processed through three layers of fully connected layers to further explore the characteristics in the graph. Finally, the classification task depends on the output produced by the final fully connected layer.

IV. EXPERIMENT

A. Experiment Setting

In light of the sample size restrictions, a stratified random choice protocol was employed. The dataset was partitioned into three distinct subsets: 70% for training, 10% for validation, and 20% for testing. Validation was integrated into each training epoch, with the parameters that maximized validation performance being preserved. These optimized parameters were subsequently applied for model testing. This cycle was iterated ten times to ensure robustness, and the resultant metrics, including their standard deviations, were meticulously documented. To systematically assess the efficacy of various methodologies, a quintet of performance metrics was deployed: accuracy, recall, precision, F1 score, and the Area Under the Receiver Operating Characteristic(ROC) Curve (AUC).

To assess the efficacy of each pattern, We carried out a contrastive study of our introduced TEMPRGL method with respect to diverse blends of partial connectivity and specific single connectivity patterns. These include: (1) Building brain networks using PC; (2) Building brain networks using SR; (3) Establishing brain networks using the Granger Causality Model(GCM); (4) Utilizing both PC and SR to build brain networks; (5) Utilizing a combination of PC and GCM for constructing brain networks; and (6) Employing a synthesis of SR and GCM for the establishment of a brain network.

B. Experimental consequence

The varying performance of our method across seven connectivity patterns is given in Table 1. Figure 2 presents the ROC curves corresponding to these methods.

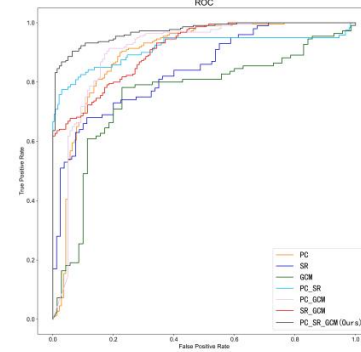


Fig 2. ROC curves of different methods

Based on these results, we can easily find that incorporating more information can significantly enhance the performance of the ASD diagnostic, which demonstrates the effectiveness of our model.

C. Ablation Analysis

To verify the effectiveness of the topological enhancement method and residual structure we used, we compared the performance results before and after adding this method, as shown in TABLE II.

Specifically, for simplicity of description, we mark the original model as Model 1, the model that only adds the residual structure is marked as Model 2, the model that only adds specific information is marked as Model 3, and our proposed model is marked as Model 4. From the results, we can easily find that our derived module can all improve the model of the original model. The results confirm that

incorporating the topological information can improve the model performance.

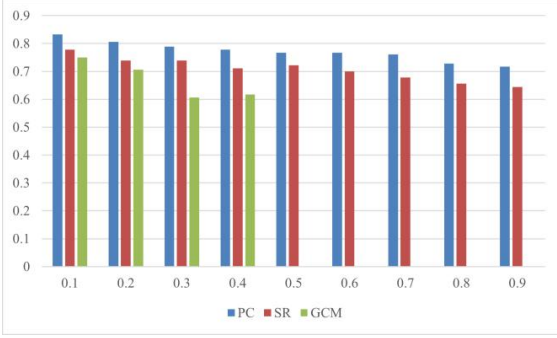


Fig 3. Performance analysis of diverse thresholds in single mode.

D. Sensitivity Analysis

To investigate the performance of estimating FBNs under different connectivity patterns and threshold levels, we conducted experiments, as shown in Fig. 3. PC, SR, and GCM achieved optimal accuracies of 83.3%, 77.8%, and 75% at a threshold of 0.1 but gradually declined with increasing thresholds. Notably, beyond a threshold of 0.4, GCM's performance dropped to zero due to its inability to construct FBNs above this level..

E. Representation of Network Characteristics

To examine the variation in FBN features acquired through different methods, we utilized the t-distributed stochastic neighbor embedding(t-SNE) algorithm to lower the dimension of FBN characteristics derived from seven various forms of connectivity into a two-dimensional depiction. The results are illustrated in Fig. 4.

It is important to note that when combining two modalities, samples of different types generally show more distinct separation than when using just one modality, while samples from the identical classification tend to cluster tightly together. This trend becomes more pronounced when incorporating three modalities. Clearly, as more connectivity modalities are combined, the observed effect becomes more significant.

F. Pos-hoc Analysis

To further validate the model's effectiveness, we conducted a post-hoc analysis using the Grad CAM method to identify key brain network connections for classification. The different network patterns are shown in Fig. 5, where the size of each sphere indicates its contribution level. Notably, Fig. 5 highlights varying contributions from brain regions related to pairwise estimation with the FBN method. Significant connections are found in the MTG, MFG, PHG,

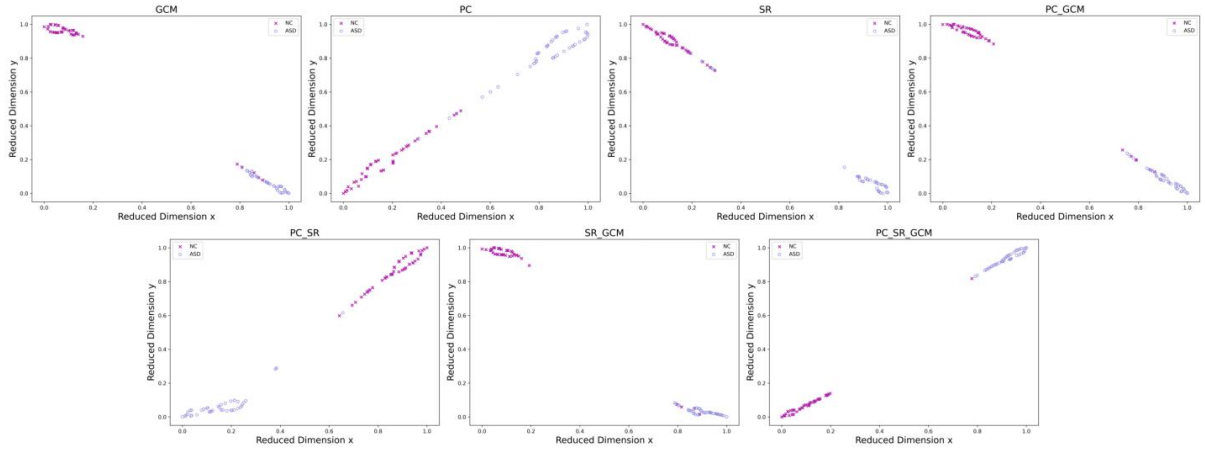
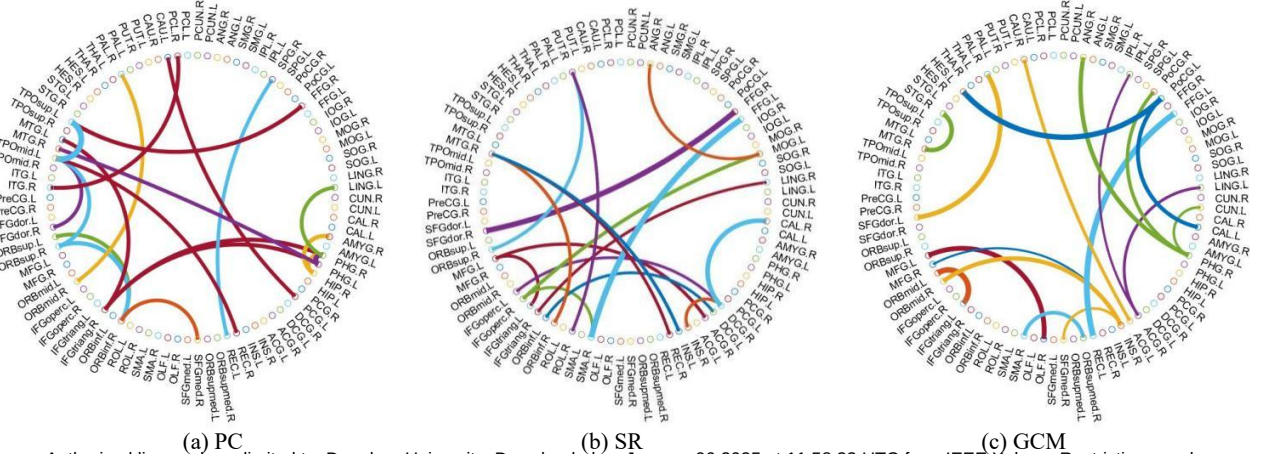


Fig 4. t-SNE visualizes feature information of different methods



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and SMA areas, all linked to ASD, illustrating our model's biological relevance. Additionally, significant differences between connection patterns emphasize the importance of integrating diverse information modalities in our model

V. CONCLUSION

In our research, we developed the Topologically Enhanced Multi-Pattern Residual Graph Learning (TEMPRGL) framework. Recognizing the limitations of current methods that focus on structural aspects of brain networks, we designed TEMPRGL to incorporate both the structural and topological information of FBNs. Our methodology involved constructing multiple brain networks from fMRI data, each representing different connectivity patterns. We then applied a dual-feature extraction strategy using multiple Residual Graph Convolutional Network (ResGCN) modules to derive multi-modal graph representations, complemented by the vectorization of FBNs. The integration of these features through our framework significantly improved the classification of neurological disorders. The experimental results not only highlight the potential of our framework in accurately diagnosing ASD but also demonstrate the importance of considering both structural and topological information in brain network analysis. The TEMPRGL framework represents a significant step forward within the context of the application of machine learning to neuroimaging data, offering an effective instrument for the early and precise neurodiagnosis disorders.

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